

# SB-GRPO: Semantic-Balanced GRPO for Enhanced LLM Reasoning

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**Abstract.** Reinforcement Learning (RL), specifically Group Relative Policy Optimization (GRPO), has emerged as a transformative technique for unlocking deep reasoning in Large Language Models (LLMs). By incentivizing multi-step Chain-of-Thought (CoT) behaviors, GRPO allows models to transcend simple pattern matching. However, current implementations are plagued by critical bottlenecks: advantage vanishing in uniform groups, systematic correction bias during reflection, and inefficient utilization of noisy, high-entropy error samples. We propose SB-GRPO (Semantic-Balanced GRPO), an advanced framework that integrates dynamic multi-layer reflection with semantic error profiling. Our architecture introduces a differentiable Heuristic Confidence Gate that smoothly transitions between standard and negative-enhanced advantage calculation, preventing gradient stalls. We further employ K-Means clustering on reasoning embeddings to identify systematic logical failure modes and construct Static Balanced Batches to stabilize policy gradients. Finally, we derive an information-theoretic loss scaling factor based on the Shannon Entropy of semantic clusters to prioritize fixing systematic errors over random noise. Evaluated under strict parameter-efficient fine-tuning constraints (4-bit QLoRA) using a 7B model, SB-GRPO achieves a highly competitive pass@1 accuracy of 74.80% on the MATH-500 dataset, yielding an absolute improvement of 26.40% over vanilla GRPO. Compared to M-GRPO, a state-of-the-art multi-layer baseline, our method delivers a 2.20% overall gain, with pronounced improvements in complex domains such as Precalculus (+5.68%) and Intermediate Algebra (+4.88%). Furthermore, on the 30-problem AIME 2024 benchmark, it improves pass@1 accuracy from 3.33% (1/30) to 10.00% (3/30).

**Keywords:** Reinforcement Learning · Large Language Models · Mathematical Reasoning · Semantic Entropy

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## 1 Introduction

The paradigm of Large Language Model (LLM) training has undergone a pivotal shift from Supervised Fine-Tuning (SFT)[16] to Reinforcement Learning-based (RL)[8] alignment. While models such as GPT-4 [1] and Claude [2] have demonstrated outstanding reasoning capabilities, their reliance on complex reward models imposes significant computational and scalability bottlenecks. Recently, the release of DeepSeek-R1 popularized Group Relative Policy Optimization (GRPO)[10] [21], marking a milestone in training specialized reasoning models without the heavy overhead of a traditional reward model. By leveraging group-relative rewards, GRPO allows the policy model to autonomously distinguish between good and bad reasoning paths within the same prompt context, effectively optimizing for complex system thinking capabilities.

However, as models scale and tasks become more complex, such as Olympiad-level mathematics and complex reasoning chains, vanilla GRPO faces diminishing returns. We identify three fundamental limitations in existing RL pipelines for reasoning:

**1. Zero Advantage:** In many reasoning tasks, especially at the start or late stages of training, a model may generate rollouts that all share the same outcome, either all correct or all incorrect. In such cases, the mean reward equals the individual reward, and the standard deviation approaches zero, causing the advantage signal to become zero [23]. This stalls the gradient and wastes expensive compute.

**2. The Compulsive Correction Fallacy:** Multi-layer [7] or iterative reflection models [14] often suffer from overcorrection. When prompted to reflect, models exhibit a systematic bias towards altering their initial answers, even when correct. This bias degrades the stability of the model outputs and leads to a high rate of false-negative revisions.

**3. The Noise-Signal Dichotomy:** Traditional RL update paradigms implicitly operate under the assumption that all optimization errors possess equivalent informational value. However, a stochastic error originating from a superficial computational oversight yields significantly less pedagogical utility than a structural error rooted in a fundamental logical misconception. Standard GRPO frameworks currently lack the granular mechanism required to disambiguate these distinct failure modes.

To address these limitations, we propose Semantic-Balanced GRPO (SB-GRPO). Our framework tackles the zero-advantage problem by introducing a Heuristic Confidence Gate, which applies a differentiable switching mechanism to activate the Advantage Calibration mechanism introduced in [15], which we formulate as a Calibrated Advantage ( $A_{calib}$ ), using virtual maximum rewards for uniform batches. To mitigate correction bias and differentiate noise from systematic errors, we integrate a Semantic Error Profiling (SEP) module within a two-layer generation-reflection pipeline. By clustering reasoning paths and constructing Static Balanced Batches, our approach ensures that the model optimizes based on the most highly informative errors rather than trivial computational noise.

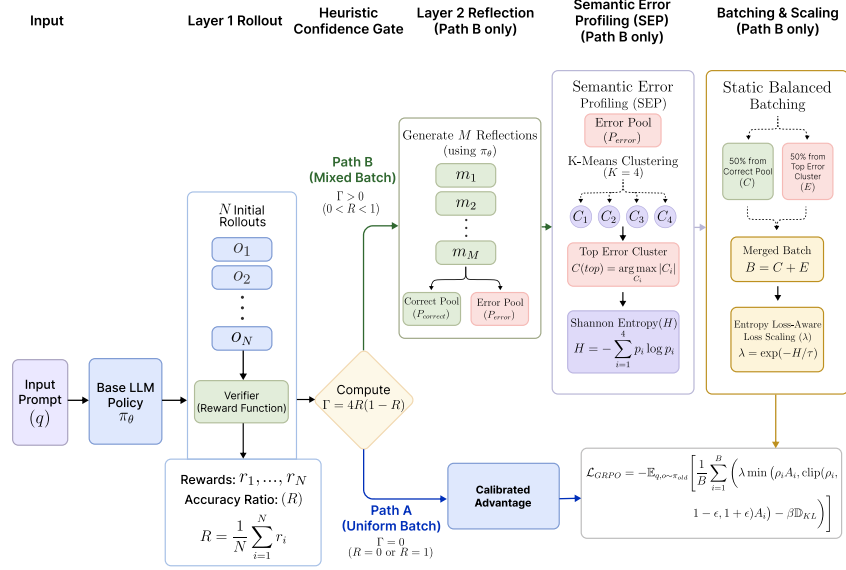
## 2 Related Work

Reinforcement Learning from Human Feedback (RLHF) [4] traditionally relies on Proximal Policy Optimization (PPO) [20], which suffers from heavy computational overhead due to separate reward models. Group Relative Policy Optimization (GRPO) [21] circumvents this bottleneck by substituting the critic with a baseline derived from the average rewards of  $G$  distinct rollouts. This memory-efficient paradigm was further refined by the Advantage Calibration mechanism introduced in NGRPO [15], which incorporates virtual maximum rewards to maintain continuous gradient flow in extreme edge cases. Concurrently, augmenting LLM reasoning via Chain-of-Thought [22] and autonomous self-correction [14] has gained significant traction. Building on this, Multi-Layer GRPO (M-GRPO) [7] employs a two-layer pipeline that utilizes initial outputs as context to verify and correct errors. However, such unconstrained self-reflective frameworks often induce a pronounced correction bias, coercing models to arbitrarily alter valid paths and thereby degrading inference stability.

To mitigate reasoning instability and overcorrection, recent studies focus on uncertainty quantification and structured data curation. Semantic Entropy (SE) [13] provides an information-theoretic metric to distinguish genuine model confusion from generative diversity—a concept SEED-GRPO [3] leverages to dynamically dampen gradient updates for highly uncertain prompts. Additionally, theoretical insights from Balanced Sampling in DIVA-GRPO [9] emphasize that maintaining an equitable distribution of correct and hard-negative samples is pivotal for stabilizing convergence in sparse-reward environments. Rather than viewing these as isolated techniques, SB-GRPO unifies and adapts these foundational concepts into an active, closed-loop multi-layer framework: dynamic gating resolves the zero-advantage bottleneck, semantic clustering extracts actionable systematic errors, and static balanced batching enforces equitable gradient signals while mitigating correction bias.

## 3 Methodology

This section details the SB-GRPO architecture (Figure 1). To resolve GRPO instability, we introduce a Heuristic Confidence Gate (Sec. 3.2) that dynamically routes uniform batches to use Advantage Calibration, preventing zero advantage. For mixed batches, Layered Generation and Reflection (Sec. 3.3) and Semantic Error Profiling (Sec. 3.4) cluster reasoning trajectories to target systematic failures. The gradients are modulated via an Information-Theoretic Loss Scaling factor (Sec. 3.5) based on error entropy. The step-by-step integration of these modules into a cohesive training iteration is formalized in Algorithm 1.



**Fig. 1.** The complete system architecture of SB-GRPO. The Heuristic Confidence Gate ( $\Gamma$ ) dynamically routes uniform batches to the Advantage Calibration module (Path A) to prevent advantage vanishing. For mixed batches (Path B), the framework applies Multi-Layer Reflection and Semantic Error Profiling (SEP) to extract the Top Error Cluster ( $C_{top}$ ), forming a Static Balanced Batch optimized by an Entropy-Aware Loss Scaling factor ( $\lambda$ ).

### 3.1 Formal Derivation of SB-GRPO

Let  $q$  be a prompt and  $O = \{o_1, \dots, o_N\}$  be  $N$  rollouts from policy  $\pi_\theta$ . The standard reward for  $o_i$  is  $r_i \in \{0, 1\}$ . In SB-GRPO, we define the accuracy ratio:

$$R = \frac{1}{N} \sum_{i=1}^N r_i$$

### 3.2 Heuristic Confidence Gate

To maintain gradient continuity, we define a differentiable heuristic gating function  $\Gamma(R)$  using the quadratic form:

$$\Gamma(R) = 4R(1 - R) \quad (1)$$

Note that  $\Gamma \in [0, 1]$ , reaching 0 at  $R = 0$  (all wrong) or  $R = 1$  (all correct), and 1 at  $R = 0.5$ . This parabolic form ensures that purely correct or incorrect batches seamlessly switch to Calibrated Advantage loss, while balanced batches

optimally utilize multi-layer reflection without gradient spikes. This gate selects the advantage calculation method:

$$A_i = \Gamma \cdot A_{\text{std},i} + (1 - \Gamma) \cdot A_{\text{calib},i} \quad (2)$$

where  $A_{\text{calib}}$  uses a virtual reward  $r_{\text{virtual}} = 1$  to inject a negative signal even when all  $r_i = 0$ . Injecting this virtual reward increases the mean  $\mu$ , which in turn pushes the  $A_{\text{std},i} = (r_i - \mu)/\sigma$  of incorrect samples into explicitly negative values rather than leaving them stuck at 0. The standard advantage is computed as  $A_{\text{std},i} = (r_i - \mu)/\sigma$ , where  $\mu$  and  $\sigma$  are the mean and standard deviation of rewards within the group. When variance approaches zero,  $A_{\text{std}}$  approaches zero, making  $A_{\text{calib}}$  critical for continuous training. Conversely, when  $R = 1$  (all correct), injecting  $r_{\text{virtual}} = 1$  maintains a zero advantage, correctly signaling that no policy update is required for an already mastered prompt.

To illustrate the practical effect of this adaptive mechanism, consider a sample size of  $N = 8$ :

- **Case 1: Homogeneous-Incorrect (0 correct, 8 wrong,  $R = 0$ ).** The gate yields  $\Gamma = 0$ . Standard GRPO would provide  $A_{\text{std}} = 0$  for all samples, halting the learning process. By routing purely to Advantage Calibration, our framework injects  $r_{\text{virtual}} = 1$ , producing an augmented mean  $\mu' \approx 0.11$  and  $\sigma' \approx 0.31$ . This assigns a meaningful negative advantage  $A \approx -0.35$  to all incorrect samples, forcing the model to explore new reasoning paths.
- **Case 2: Low-Accuracy Mixed (1 correct, 7 wrong,  $R = 0.125$ ).** The gate yields  $\Gamma = 0.4375$ . Standard GRPO assigns an overly aggressive reward to the single correct sample ( $A_{\text{std}} \approx +2.65$ ) and a weak penalty to the wrong ones ( $A_{\text{std}} \approx -0.38$ ). Our dynamically routed advantage rebalances these to  $A_{\text{correct}} \approx +2.21$  and  $A_{\text{wrong}} \approx -0.47$ . This dampens the variance of lucky guesses while increasing the penalty for failures, strongly promoting exploration when accuracy is low.
- **Case 3: High-Accuracy Mixed (7 correct, 1 wrong,  $R = 0.875$ ).** The gate yields  $\Gamma = 0.4375$ . Standard GRPO computes  $A_{\text{std}} \approx +0.38$  for correct and  $-2.65$  for incorrect samples. Our routed advantage minimally shifts these to  $+0.36$  and  $-2.75$ , respectively. In this high-accuracy scenario, the mechanism makes minimal adjustments, stably favoring exploitation while maintaining a strict penalty for the rare failure.

### 3.3 Layered Generation and Reflection

Rather than relying on a single-pass generation, SB-GRPO isolates the exploration and verification phases into a two-layer pipeline. This deliberate separation allows the model to systematically audit its own reasoning paths before gradient updates are computed. The process is defined as follows:

**Layer 1 (Rollout):**  $N = 8$  samples are generated with temperature  $\tau = 0.7$  to explore the solution space.

**Algorithm 1** SB-GRPO Training Step

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1: Input: Prompt  $q$ , Policy  $\pi_\theta$ , Verifier  $V$ 
2: Generate  $N$  rollouts  $O = \{o_1, \dots, o_N\}$  via vLLM
3:  $R \leftarrow \text{Accuracy}(O, V)$ 
4:  $\Gamma \leftarrow 4R(1 - R)$ 
5: if  $\Gamma > 0$  then
6:   Generate  $M$  reflections per rollout  $\rightarrow$  Pool  $P$ 
7:   Cluster  $P_{\text{error}}$  into  $\{C_1 \dots C_K\}$  using K-Means
8:    $\lambda \leftarrow \exp(-\text{Entropy}(C)/\tau)$ 
9:    $B \leftarrow \text{Sample } S/2 \text{ from } P_{\text{correct}} \text{ and } S/2 \text{ from } C_{\text{top}} \text{ (with replacement)}$ 
10: else
11:    $\lambda \leftarrow 1.0$ ,  $B \leftarrow \text{Sample } S \text{ uniformly from } O \text{ (with replacement)}$ 
12:   Use Calibrated Advantage  $A_{\text{calib}}$ 
13: end if
14: Training Phase:
15: Flush vLLM cache  $\rightarrow$  Init PyTorch Trainer
16:  $\rho \leftarrow \frac{\pi_\theta}{\pi_{\text{old}}}$ 
17:  $\mathcal{L}_{\text{surr}} \leftarrow \min(\rho A, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon) A)$ 
18:  $\mathcal{L}_{\text{GRPO}} \leftarrow \frac{1}{|B|} \sum (-\lambda \mathcal{L}_{\text{surr}} + \beta \mathbb{D}_{KL})$ 
19: Backward and Optimizer Step

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**Layer 2 (Reflection):** If  $\Gamma > 0$ , the model performs  $M = 2$  reflections per rollout using a neutral system prompt: *"Review the solution logic. Only correct it if errors are found; otherwise, maintain consistency."* This creates a pool of  $N \times M = 16$  reasoning paths.

### 3.4 Semantic Error Profiling (SEP)

To effectively disambiguate structural reasoning flaws from superficial generation noise, the SEP module projects all incorrect trajectories into a dense semantic space. Specifically, incorrect samples are vectorized utilizing a frozen *all-MiniLM-L6-v2* encoder [17]. Subsequently, K-Means clustering [19] (empirically set to  $K = 4$ ) is applied to partition these latent embeddings into a discrete set of semantic clusters  $C = \{C_1, \dots, C_K\}$ . By identifying the top error cluster, defined mathematically as  $C_{\text{top}} = \arg \max_k |C_k|$ , we isolate the most prevalent logical failure mode associated with the given prompt. This isolation provides a concentrated learning signal for targeted error correction, ensuring the policy’s optimization against systematic misconceptions rather than stochastic, trivial errors.

### 3.5 Information-Theoretic Loss Scaling

To mathematically quantify the epistemic uncertainty inherent in the model’s failure modes, we compute the Shannon Entropy  $H$  over the semantic cluster distribution. Let  $p_k = |C_k| / \sum_{j=1}^K |C_j|$  denote the probability mass of the  $k$ -th

cluster. The entropy is defined as:

$$H = - \sum_{k=1}^K p_k \log(p_k) \quad (3)$$

Based on this uncertainty metric, we introduce a dynamic loss scaling factor  $\lambda = \exp(-H/\tau)$ , where  $\tau$  serves as a temperature hyperparameter (set empirically to  $\tau = 1.0$ ). This scaling factor is directly applied to modulate the policy gradient. Intuitively, a high entropy value signifies a highly dispersed, random distribution of superficial errors (e.g., stochastic arithmetic mistakes), thereby dampening the gradient update to prevent the model from overfitting to noise. Conversely, a low entropy value indicates a concentrated, systematic logical flaw, triggering a remarkably larger, more aggressive optimization step to rectify the foundational misunderstanding.

### 3.6 Semantic-Balanced GRPO Objective Function

The final optimization objective integrates the standard PPO clipped surrogate loss, the KL divergence penalty, and our novel entropy-aware loss scaling factor  $\lambda$ . For a prompt  $q$  and a static balanced batch  $B$  of size  $S$ , the policy  $\pi_\theta$  is updated by maximizing the objective:

$$\mathcal{J}_{\text{SB-GRPO}}(\theta) = \frac{1}{S} \sum_{i=1}^S \left[ \lambda \min \left( \frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)} A_i, \text{clip} \left( \frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \right] \quad (4)$$

where  $A_i$  is the dynamically routed advantage (via the Heuristic Confidence Gate  $\Gamma$ ),  $\epsilon$  is the clipping hyperparameter, and  $\beta$  controls the KL penalty against the reference model  $\pi_{\text{ref}}$ . By scaling the surrogate objective with  $\lambda$ , the magnitude of the policy update is strictly regulated by the semantic concentration of the error clusters, forcing the model to heavily penalize and unlearn systematic logical flaws while conservatively adjusting for noisy, random errors.

## 4 Datasets and Experimental Setup

### 4.1 Datasets

To comprehensively evaluate the reasoning capabilities and self-correction effectiveness of our proposed approach, we utilize a combination of well-established mathematical reasoning datasets for both training and evaluation.

**Training Data:** Our training dataset is constructed by systematically aggregating math-focused data: **MATH (Levels 3–5)** [11] (5,586 examples filtered to exclude trivial tiers, ensuring multi-step challenge) and **GSM8K** [5] (7,473 standard training problems), providing a robust foundation of foundational arithmetic and advanced algebraic problem-solving skills.

**Evaluation Benchmarks:** For evaluation, we employ rigorous benchmarks: **MATH-500** (500 curated problems from the MATH test split), **AIME 2024 & 2025** (30 challenging problems each from the American Invitational Mathematics Examination), **GPQA Diamond** [18] (198 PhD-level science questions), and **GSM8K** (1,319 test questions assessing arithmetic consistency without correction bias).

## 4.2 Experimental Setup

All experiments, including the training of our SB-GRPO models and all baseline comparisons (Vanilla GRPO, M-GRPO), are conducted under a unified, strictly controlled environment to ensure fair comparisons. We adopt **Qwen/Qwen2.5-7B-Instruct** as the foundational base model for all configurations. Due to memory constraints when concurrently running a 7B model and a generation engine, we apply 4-bit QLoRA [12,6] (`load_in_4bit=True`, `use_peft=True`) alongside DeepSpeed ZeRO-3 and Flash Attention 2 to optimize memory efficiency and computation speed. All training runs are executed on a single high-performance Linux server equipped with one NVIDIA H100 GPU (80GB VRAM). We train the models for 1 epoch with a learning rate of  $1.0 \times 10^{-5}$ . To maintain consistent memory usage, the maximum prompt length is constrained to 2048 tokens, and the completion length is capped at 1024 tokens. For each prompt, the model generates  $N = 8$  independent completions to construct the reasoning pool.

**Evaluation Protocol:** To evaluate both peak deterministic performance and variance on small discrete test suites (e.g., AIME’s 30 problems), we report both single-sample greedy decoding (Table 1) and multi-pass evaluation (Table 2) averaged over 4 independent evaluation passes with distinct generation seeds on the trained policy checkpoint.

**Hardware-Aware Static Tensor Shaping:** To maximize GPU throughput on the NVIDIA H100 and completely mitigate memory fragmentation induced by variable sequence lengths, we strictly avoid dynamic padding. Building upon the variance reduction principles of DIVA-GRPO, we construct a Static Balanced Batch of a fixed size  $S = 16$ . Specifically, we sample  $S/2$  trajectories from the Correct Pool and the remaining  $S/2$  from the Top Error Cluster ( $C_{\text{top}}$ ). By enforcing Sampling with Replacement, we guarantee that the tensor shape remains entirely constant across all training steps. This static geometric configuration minimizes memory allocation overhead, thereby fully maximizing the computational efficiency of the DeepSpeed ZeRO-3 backend under constrained VRAM environments.

## 5 Experimental Results

### 5.1 Performance Evaluation

All models were evaluated using the `lighteval` framework on the exact same hardware configuration to ensure fairness. Table 1 compares SB-GRPO with vanilla GRPO (which lacks reflection entirely) and M-GRPO [7]. M-GRPO is chosen as our primary baseline because it represents the state-of-the-art in multi-layer reinforcement learning for self-correction. By utilizing the same two-layer reflection structure but sampling revisions uniformly without semantic clustering or balancing, M-GRPO serves as a direct control group (ablation). This allows us to isolate the performance gains driven specifically by our semantic profiling and balanced batching mechanisms, minimizing architectural confounding variables.

**Table 1.** Pass@1 Accuracy (%) and absolute performance gains ( $\Delta$ ) on Advanced Reasoning Benchmarks (1 Sample Greedy Decoding).  $\Delta_{S-V}$  and  $\Delta_{S-M}$  denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

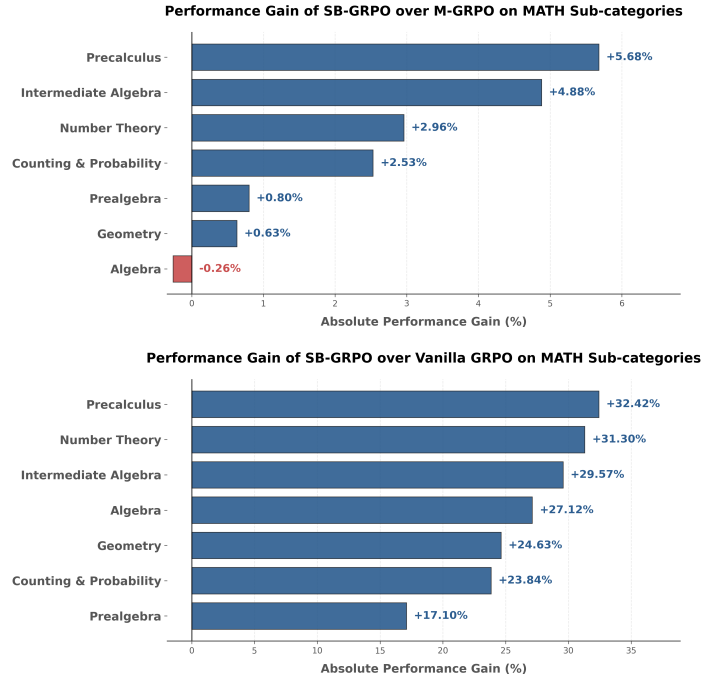
| Dataset      | Accuracy (%) |              |                | Absolute Gains (%) |                |
|--------------|--------------|--------------|----------------|--------------------|----------------|
|              | Vanilla      | M-GRPO       | <b>SB-GRPO</b> | $\Delta_{S-V}$     | $\Delta_{S-M}$ |
| MATH-500     | 48.40        | 72.60        | <b>74.80</b>   | <b>26.40</b>       | <b>2.20</b>    |
| AIME 2024    | 3.33         | 6.67         | <b>10.00</b>   | <b>6.67</b>        | <b>3.33</b>    |
| AIME 2025    | 3.33         | <b>6.67</b>  | <b>6.67</b>    | <b>3.34</b>        | 0.00           |
| GPQA Diamond | 30.30        | 31.82        | <b>32.83</b>   | <b>2.53</b>        | <b>1.01</b>    |
| GSM8K        | 56.33        | <b>70.51</b> | 69.22          | <b>12.89</b>       | -1.29          |

**Table 2.** Pass@1 Accuracy (%) and absolute performance gains ( $\Delta$ ) on advanced reasoning benchmarks evaluated with 4 generation passes (mean  $\pm$  stderr from `lighteval`).  $\Delta_{S-V}$  and  $\Delta_{S-M}$  denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

| Dataset      | Accuracy (% $\pm$ stderr) |                  |                                    | Absolute Gains (%) |                |
|--------------|---------------------------|------------------|------------------------------------|--------------------|----------------|
|              | Vanilla                   | M-GRPO           | <b>SB-GRPO</b>                     | $\Delta_{S-V}$     | $\Delta_{S-M}$ |
| MATH-500     | 47.60 $\pm$ 1.85          | 72.30 $\pm$ 1.68 | <b>75.65 <math>\pm</math> 1.65</b> | <b>+28.05</b>      | <b>+3.35</b>   |
| AIME 2024    | 1.67 $\pm$ 1.67           | 10.83 $\pm$ 4.59 | <b>12.50 <math>\pm</math> 4.91</b> | <b>+10.83</b>      | <b>+1.67</b>   |
| AIME 2025    | 0.83 $\pm$ 0.83           | 6.67 $\pm$ 3.38  | <b>8.33 <math>\pm</math> 3.46</b>  | <b>+7.50</b>       | <b>+1.66</b>   |
| GPQA Diamond | 28.91 $\pm$ 2.17          | 29.42 $\pm$ 2.36 | <b>33.08 <math>\pm</math> 2.45</b> | <b>+4.17</b>       | <b>+3.66</b>   |

### 5.2 Analysis

The results demonstrate that SB-GRPO’s core strength lies in resolving high-complexity reasoning tasks:



**Fig. 2.** Absolute performance gains of SB-GRPO on MATH sub-categories. **Top:** Compared to M-GRPO, showing pronounced improvements in *Intermediate Algebra* (+4.88%) and *Precalculus* (+5.68%). **Bottom:** Compared to Vanilla GRPO, highlighting broad gains achieved by multi-layer reasoning.

**Table 3.** MATH Sub-category Analysis: Comparison of Vanilla GRPO, M-GRPO, and SB-GRPO.  $\Delta_{S-V}$  and  $\Delta_{S-M}$  denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

| Category               | Accuracy (%) |              |                | Absolute Gains (%) |                |
|------------------------|--------------|--------------|----------------|--------------------|----------------|
|                        | Vanilla      | M-GRPO       | <b>SB-GRPO</b> | $\Delta_{S-V}$     | $\Delta_{S-M}$ |
| Algebra                | 62.01        | <b>89.39</b> | 89.13          | <b>27.12</b>       | -0.26          |
| Counting & Probability | 43.25        | 64.56        | <b>67.09</b>   | <b>23.84</b>       | <b>2.53</b>    |
| Geometry               | 32.99        | 56.99        | <b>57.62</b>   | <b>24.63</b>       | <b>0.63</b>    |
| Intermediate Algebra   | 25.14        | 49.83        | <b>54.71</b>   | <b>29.57</b>       | <b>4.88</b>    |
| Number Theory          | 42.22        | 70.56        | <b>73.52</b>   | <b>31.30</b>       | <b>2.96</b>    |
| Prealgebra             | 65.33        | 81.63        | <b>82.43</b>   | <b>17.10</b>       | <b>0.80</b>    |
| Precalculus            | 24.54        | 51.28        | <b>56.96</b>   | <b>32.42</b>       | <b>5.68</b>    |
| <b>Overall</b>         | 42.21        | 66.32        | <b>68.78</b>   | <b>26.57</b>       | <b>2.46</b>    |

**Single-Sample Evaluation (Table 1):** Under 1-sample greedy decoding, SB-GRPO achieves +2.20% on MATH-500, +3.33% on AIME 2024, and +1.01% on GPQA Diamond over M-GRPO. On AIME 2025, SB-GRPO achieves 6.67%, performing on par with M-GRPO ( $\Delta_{S-M} = 0.00\%$ ) in this single-sample snapshot. Because AIME comprises only 30 problems, single-pass evaluations exhibit discrete quantization granularity ( $\pm 3.33\%$  per problem), where a difference of a single question dictates the metric. This directly motivated our multi-pass evaluation protocol to capture true generative consistency.

**Multi-Sample Evaluation (Table 2):** Across multi-sample generation (4 passes per problem) with distinct seeds on the trained policy, the advantages of semantic balancing become statistically robust. Table 2 reports Pass@1 accuracy alongside standard errors (mean  $\pm$  stderr) directly computed by the `lighteval` evaluation framework. SB-GRPO achieves  $75.65 \pm 1.65\%$  on MATH-500 (+3.35% over M-GRPO) and  $33.08 \pm 2.45\%$  on GPQA Diamond (+3.66%). Crucially, on AIME 2025, multi-pass evaluation breaks the single-sample tie, with SB-GRPO achieving  $8.33 \pm 3.46\%$  (10/120 solved) vs.  $6.67 \pm 3.38\%$  (8/120 solved) for M-GRPO. On AIME 2024, SB-GRPO achieves  $12.50 \pm 4.91\%$  (15/120) compared to  $10.83 \pm 4.59\%$  (13/120) for M-GRPO and  $1.67 \pm 1.67\%$  (2/120) for Vanilla GRPO. This multi-sample protocol confirms that gains are consistent across diverse generation seeds rather than stochastic single-pass anomalies.

**Sub-category Analysis (Table 3):** As depicted in Fig. 2, SB-GRPO exhibits its most substantial advantages in rigorous multi-stage disciplines: Precalculus (+5.68%) and Intermediate Algebra (+4.88%). For saturated domains like elementary Algebra ( $\sim 89\%$  accuracy), failure modes are predominantly sporadic arithmetic slips rather than structural logical flaws; the entropy loss scaling factor adaptively downscales updates to preserve policy representations, yielding performance parity (89.13% vs 89.39%, -0.26%).

### 5.3 Ablation Analysis: Deconstructing the Performance Gains

To understand how individual components contribute to the final gains, we systematically deconstruct the three core mechanisms of SB-GRPO and examine their operational effects:

**1. Role of Heuristic Confidence Gate ( $\Gamma$ ) and Advantage Calibration:** In standard GRPO, whenever a rollout group is uniform ( $R = 0$  or  $R = 1$ ), reward variance vanishes ( $\sigma \rightarrow 0$ ), reducing advantage values to zero ( $A_{\text{std}} = 0$ ) and stalling policy gradients entirely (Case 1, Sec. 3.2). Ablating  $\Gamma$  leads to repeated zero-gradient updates on difficult prompts. The gate smoothly activates virtual-reward calibration ( $A_{\text{calib}}$ ) for uniform batches, sustaining continuous gradient flow without inducing optimization spikes.

**2. Role of Semantic Error Profiling (SEP) and Static Balanced Batching:** M-GRPO represents the structural ablation where two-layer reflection is enabled, but revisions are sampled uniformly without semantic clustering or balance. As shown in Table 1, naive reflection plateaus at 72.60% on MATH-500. Replacing uniform sampling with SEP and static 50/50 balanced batches ( $S = 16$ ) concentrates gradient descent on the dominant error cluster ( $C_{\text{top}}$ ) while maintaining parity with verified reasoning paths, delivering an immediate +2.20% gain.

**3. Role of Information-Theoretic Loss Scaling ( $\lambda$ ):** Without entropy loss scaling ( $\lambda = 1.0$ ), the policy treats all errors with equal update magnitude. The GSM8K results (Sec. 5.4) illustrate why this is sub-optimal: simple arithmetic slips produce high semantic entropy across clusters, where  $\lambda = \exp(-H/\tau)$  adaptively attenuates updates to prevent overfitting to superficial noise. When errors concentrate into a systematic fallacy (low  $H$ ),  $\lambda$  scales up to enforce decisive error unlearning.

**4. Structural Mitigation of Correction Bias in Self-Reflection:** A primary motivation of SB-GRPO is counteracting the *Compulsive Correction Fallacy* inherent to unguided multi-layer reflection. In standard reflective GRPO (M-GRPO), prompting reflection unconditionally often induces models to second-guess valid intermediate deductions, converting initially correct solutions into incorrect revisions (false-negative drift). SB-GRPO structurally counters this vulnerability through two design choices: (i) an objective, neutral reflection prompt (Sec. 3.3) that prevents biasing the policy toward unwarranted modifications, and (ii) Static Balanced Batching ( $S/2$  verified trajectories paired with  $S/2$  from  $C_{\text{top}}$ ), preventing gradient updates from being skewed by negative paths. This architectural pairing stabilizes correct reasoning paths while strictly targeting persistent, clustered logical flaws.

### 5.4 Discussion

**The GSM8K Trade-Off:** SB-GRPO scores 69.22% on GSM8K compared to M-GRPO’s 70.51%. GSM8K consists primarily of elementary word problems with short reasoning chains, where incorrect answers stem from stochastic calculation slips rather than structural logical flaws. Consequently, semantic entropy

$H$  remains uniformly high, prompting  $\lambda$  to dampen gradient updates. While M-GRPO overfits to these arithmetic noise samples, SB-GRPO preserves representational stability, translating into decisive superiority on advanced benchmarks.

**Qualitative Inspection of Semantic Error Clusters:** Qualitative inspection of the Top Error Cluster ( $C_{\text{top}}$ ) confirms our core premise: on Intermediate Algebra and Precalculus problems in MATH-500, incorrect rollouts within  $C_{\text{top}}$  consistently share identical mathematical misconceptions—such as misapplying sign reversals in quadratic inequalities or neglecting boundary conditions in logarithmic domains—rather than arbitrary lexical overlap. This demonstrates that frozen MiniLM embeddings reliably isolate systematic logical flaws.

## 6 Conclusion and Future Work

In this paper, we introduced Semantic-Balanced Group Relative Policy Optimization (SB-GRPO), a reinforcement learning framework designed to overcome the critical bottlenecks of advantage vanishing and correction bias in LLM reasoning. By synthesizing a Heuristic Confidence Gate with Semantic Error Profiling and an Information-Theoretic Loss Scaling factor, SB-GRPO dynamically stabilizes gradients across highly skewed batches while actively targeting systematic logical flaws. Empirical evaluations demonstrate that this architecture achieves competitive performance on complex mathematical benchmarks, achieving 10.00% on AIME 2024 and yielding substantial improvements on the MATH-500 dataset under strict hardware and parameter constraints (4-bit QLoRA [12,6]).

Notably, SB-GRPO establishes an efficient paradigm for aligning reasoning models, demonstrating that targeted semantic error curation surpasses unguided reflection. Sub-category analysis highlights that SB-GRPO excels in high-complexity reasoning, specifically in *Precalculus* and *Intermediate Algebra*, significantly outperforming existing baselines. Future work will explore extending this entropy-aware balancing mechanism to multi-modal reasoning tasks and evaluating its cross-architecture generalizability across diverse foundation model backbones beyond Qwen2.5-7B.

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